

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

Year/Sem: III / II

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Lab Objectives

- Apply advanced prompt engineering in real-world scenarios
- Use AI to solve classification and recommendation problems
- Analyze AI-generated outputs
- Improve prompts for better accuracy

Task 1: Email Classification

Prompt: Classify an email as Spam or Not Spam.

Python Code

```
email = input("Enter email content: ")

spam_keywords = ["win", "prize", "free", "money"]

if any(word in email.lower() for word in spam_keywords):
    print("Spam")
else:
    print("Not Spam")
```

Task 2: Chatbot Intent Detection

Prompt: Detect user intent from input message.

Python Code

```
msg = input("Enter message: ")

if "book" in msg:
    print("Intent: Booking")
elif "cancel" in msg:
    print("Intent: Cancellation")
else:
    print("Intent: General Query")
```

Task 3: Feedback Analysis

Prompt: Analyze whether feedback is Positive or Negative.

Python Code

```
feedback = input("Enter feedback: ")

if "good" in feedback.lower() or "excellent" in feedback.lower():
    print("Positive")
else:
    print("Negative")
```

Task 4: Course Recommendation

Prompt: Recommend a course based on user interest.

Python Code

```
interest = input("Enter interest: ")

if "data" in interest.lower():
    print("Recommended: Data Science")
elif "web" in interest.lower():
    print("Recommended: Web Development")
else:
    print("Recommended: Programming Fundamentals")
```

Conclusion

AI prompt engineering can be applied to real-world applications such as classification, recommendation, and chatbot systems.

Well-structured prompts improve the quality and accuracy of results.